



Digital Technologies Enabling Green Supply Chain Management Practices in Vietnam's Electronics Industry

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Abstract: This study examines how digital technologies shape green supply chain management (GSCM) in Vietnam's electronics industry. Using an exploratory qualitative multi-case design, we investigated two leading Vietnamese electronics firms and triangulated evidence from company documents, field observations and 12 semi-structured interviews, including 10 interviews across the two focal firms and two interviews with external experts. Interviewees represented senior management and key functions related to environmental management, production, procurement and technology, with interview duration ranging from 80 to 120 minutes. The cases suggest an internal environmental management-led digitalization pattern in which firms first deploy digital tools for internal environmental monitoring and control and subsequently strengthen greener manufacturing and environmental cooperation, while green procurement and reverse logistics tend to lag when data integration, supplier participation and analytics capabilities remain limited. Digital adoption appears to support operational efficiency, environmental performance, employee capability development, supplier participation and faster Environmental, Social, and Governance (ESG) compliance responses, enabled by real-time sustainability information and reuse practices linked to reverse logistics. However, implementation is constrained by investment costs, skills gaps, fragmented systems and cybersecurity risks, reinforced by uncertain sustainability requirements and weak domestic green demand. The findings provide exploratory multi-case evidence from two leading Vietnamese electronics firms on practice-specific digitalization in an emerging-economy context and propose a staged digital-green capability-building roadmap. The study supports analytical rather than statistical generalization and should be interpreted as theory-building evidence for digital-green supply-chain transformation.

Keywords: Artificial Intelligence; Internet of Things; Big data; Digital technologies; Electronics industry; Green supply chain management; Sustainability; Vietnam

1. Introduction

In the context of climate change and increasing sustainability pressures, green supply chain management (GSCM) has become a strategic priority for global businesses (Hariyani et al., 2024; Khan et al., 2025). In emerging economies, GSCM is often realized through five core green supply chain management practices (GSCMPs), including internal environmental management (IEM), green procurement (GP), green manufacturing (GM), reverse logistics (RL), and environmental cooperation (EC), which are used to structure environmental measurement and implementation in enterprises (Balon, 2020; Zhu & Sarkis, 2006; Zhu et al., 2008).

In addition, the Industrial Revolution 4.0 with digital technologies such as the Internet of Things (IoT), big data (BD) and Artificial Intelligence (AI) is fundamentally changing the supply chain through improving connectivity, integration, and chain performance (Rad et al., 2022; Srhir et al., 2023). IoT enables real-time monitoring, increased transparency, and traceability along the chain with track/trace platforms and real-time data (Taj et al., 2023); BD helps with demand analysis and inventory optimization; while AI aids in forecasting, production

automation, and waste reduction (Benzidia et al., 2021). Studies have shown that digitalization has a positive impact on green supply chain integration, reducing costs, and improving ESG performance (Akbari & Hopkins, 2022; Benzidia et al., 2021).

Vietnam's electronics industry was chosen as the research context because it is a key export-oriented sector, contributing substantially to national exports and industrial production. However, its capacity to participate in global green supply chains remains constrained by dependence on imported components, foreign-invested production networks, low domestic value added, and uneven supporting-industry capability (ILO, 2023; MOIT, 2024; Vietnam Briefing, 2025; Vietnam News, 2025). This creates an urgent need to examine how domestic enterprises use IoT, BD and AI to overcome digital-green transformation challenges and improve competitiveness.

Although studies have documented the positive impact of digitalization on GSCM, much of the research has focused on an individual practice or context of developed economies; qualitative, multi-practice evidence for the electronics industry in emerging economies is scarce (Oyedijo et al., 2024; Qiao et al., 2023). This study fills that gap by designing a multi-case study at two large-scale electronics enterprises in Vietnam, following established logic for theory building and replication in case-study research (Eisenhardt, 1989; Voss et al., 2016; Yin, 2018) with 3 main research questions:

To sharpen the theoretical focus early in the paper, this study conceptualizes digitalization in GSCMPs as a staged capability-building process rather than a simple technology-adoption outcome. We argue that IoT, BD and AI first create internal environmental data visibility, particularly through internal environmental management (IEM). This internal visibility then becomes a foundation for higher-order digital-green capabilities, including integrated data governance, predictive analytics, supplier evaluation, reverse-logistics tracking and inter-organizational environmental collaboration. Accordingly, the study examines a possible IEM-led digitalization pattern in which firms first digitalize internal environmental monitoring and control before extending digital capabilities to GM, EC, GP and RL. This pattern is treated as an interpretive mechanism derived from the two cases rather than as a generally verified causal effect.

RQ1. What is the level and current status of application of digital technologies (IoT, BD, AI) in each GSCMP (IEM, GP, GM, RL, EC) at the two Vietnamese electronics companies today?

RQ2. What are the specific benefits of the application of digital technologies, and what barriers are companies facing in the process of implementing GSCMPs?

RQ3. How do the characteristics of Vietnam's electronics industry affect the application of digital technologies in GSCMPs?

This study contributes on both theoretical and practical levels, but its contributions should be interpreted as exploratory and analytical rather than statistically generalizable. First, it develops an integrated conceptual framework that connects institutional pressures, TOE adoption conditions, RBV-based digital-green resources and dynamic capabilities to explain how digital technologies may diffuse across GSCMPs. Second, it provides exploratory qualitative multi-case evidence from two leading Vietnamese electronics firms, an emerging-economy context where ESG requirements, supplier dependence, data fragmentation and weak domestic green demand shape digital-green transformation. Third, it proposes an observed IEM-led digitalization pattern and suggests that data governance may help convert digital technologies into practice-specific GSCMP improvements. Practically, the findings help electronics firms and policymakers identify benefits, barriers and staged implementation priorities when integrating IoT, BD and AI into GSCMPs, while recognizing the boundary conditions of large-firm cases and the need for further validation.

2. Literature Review

2.1 Green Supply Chain Management Practices Common in Emerging Economies

GSCM integrates environmental, social and economic objectives throughout the supply chain, from raw material extraction to end-of-life product processing, towards sustainable development (Ali et al., 2017; Seuring & Müller, 2008). The concept emerges as a business response to regulatory compliance pressures and global sustainability requirements (Sarin & Srivastava, 2024). In emerging economies, GSCM is increasingly popular, both helping to improve operational efficiency and meeting international standards (Ali et al., 2017).

Sustainability thinking in supply-chain research has evolved from a narrow environmental-compliance orientation toward a broader sustainability-transition perspective. Early GSCM studies mainly emphasized pollution prevention, cleaner production, regulatory compliance and environmental performance. Later sustainable supply-chain management extended this view by integrating environmental, economic and social objectives across supply-chain activities. More recent circular and digitally enabled perspectives further emphasize lifecycle thinking, reverse logistics, resource recovery, supplier collaboration, data transparency and resilience. In this trajectory, digital technologies are not only operational tools; they also make sustainability information more visible, measurable and actionable across organizational boundaries (Islam et al., 2024; Zighan & Ruel, 2025).

In this study, environmental performance refers specifically to outcomes such as energy reduction, emission

reduction, waste reduction, resource efficiency, reuse and recycling. Sustainability performance refers to the broader triple-bottom-line outcomes covering environmental, economic and social dimensions. ESG is used more narrowly to refer to reporting, disclosure, compliance and governance systems that manage environmental, social and governance information. This distinction is used consistently in the revised analysis to avoid treating ESG, environmental performance and sustainability performance as interchangeable concepts.

The five core GSCMPs used in this study, confirmed in prior studies by Balon (2020), Zhu & Sarkis (2006) and Zhu et al. (2008), include: (i) IEM, referring to leadership commitment and internal environmental management systems such as ISO 14001; (ii) GP, referring to the integration of ecological criteria into supplier selection and purchasing decisions (Nazir et al., 2024); (iii) GM, referring to production and process design that reduce energy use, chemical consumption and waste (Seuring & Müller, 2008); (iv) RL, referring to product recovery, reuse and recycling activities that help close the supply-chain loop (Zhu et al., 2008); and (v) EC, referring to information sharing and collaboration with stakeholders to implement green initiatives (Balon, 2020). This framework provides the foundation for connecting external pressures, internal capabilities and sustainability-related performance outcomes (Balon, 2020; Islam et al., 2024).

The ability to implement GSCMPs depends on external pressure and internal capacity. Typical external pressures such as requirements to comply with environmental laws and regulations, pressure from international customers and partners are considered important drivers for businesses to implement GSCM (Balon, 2020; Nazir et al., 2024). In Vietnam, international regulations (RoHS, WEEE, REACH) have forced many electronics manufacturers to adjust their processes in a greener direction (Tsai et al., 2023). In addition, the commitment of leadership, organizational culture, and financial and human resources determine the level of implementation of GSCMPs (Ali et al., 2017). In particular, the level of digital technologies application is emerging as a key factor positively affecting the implementation of GSCMPs: many recent studies confirm the application of digital technologies such as IoT, BD, AI significantly facilitate green supply chain governance, thereby simultaneously improving environmental and economic efficiency (Akbari & Hopkins, 2022; Ghadge et al., 2022; Qiao et al., 2023).

To explain this mechanism, this study does not treat institutional theory, RBV, TOE and dynamic capabilities as separate lenses; rather, it integrates them into a staged analytical logic. Institutional theory explains why firms face legal, market and normative sustainability pressures from regulators, multinational customers and industry standards (DiMaggio & Powell, 1983; Zhu & Sarkis, 2006). The TOE framework explains the conditions under which firms adopt green digital technologies, including technological readiness, organizational resources and environmental pressure (Sarin & Srivastava, 2024). RBV explains why IoT sensors, BD platforms, AI tools, environmental-management systems and data infrastructure become valuable resources only when embedded in firm-specific capabilities such as data governance, employee skills and supplier collaboration (Barney, 1991). Dynamic capabilities then explain how firms sense sustainability requirements, seize digital opportunities and transform routines to diffuse digital technologies across GSCMPs (Wamba et al., 2024; Li & Lin, 2024; Teece, 2007).

In Vietnam, the implementation of GSCMPs is still in its infancy. Many electronics firms still stop at complying with mandatory regulations or participating in some environmental programs to meet global supply chains (Tsai et al., 2023). However, financial, human and technological barriers make it difficult for GSCMPs to spread widely. In this context, the application of digital technologies has become an urgent direction, opening up opportunities to overcome limitations and improve sustainable competitiveness.

2.2 Application of Digital Technology in Green Supply Chain Management Practices in Emerging Economies

Adoption patterns in emerging economies. In emerging economies, the integration of digital technology into GSCMPs remains uneven and generally slower than in developed countries (Sarin & Srivastava, 2024). In Vietnam, available survey evidence suggests that IoT is the most widely applied technology, while AI and BD analytics remain at an early stage of adoption (Akbari & Hopkins, 2022). Evidence from other emerging economies also shows that Industry 4.0 diffusion varies substantially across countries because of differences in infrastructure, human resources, policy support and firm-level capabilities (Alshahrani, 2023). Overall, emerging economies are attempting to catch up with supply-chain digitalization, but adoption remains conditioned by internal capabilities and external integration pressures.

Cross-industry applications. Across sectors, digital technologies are applied to GSCM in different ways depending on product characteristics, traceability requirements and market expectations. In textile and fashion supply chains, IoT and blockchain can support material traceability, water-use monitoring and recycling transparency (Nishan et al., 2024). In retail and consumer-goods supply chains, AI and BD improve demand forecasting and inventory management, thereby reducing excess stock and waste (Hübner et al., 2024). In logistics, IoT-based tracking and AI-enabled route optimization improve transport visibility and may reduce fuel consumption and emissions (Sharma et al., 2023). In food supply chains, blockchain can enhance farm-to-fork

traceability and support the removal of non-compliant products (Bhawna et al., 2024).

These cross-industry examples show that digital technologies can support environmental transparency and operational efficiency, but their effects depend on industry-specific product characteristics, data requirements and market pressures. This is particularly relevant for the electronics industry, where short product life cycles, complex components, e-waste risks and multi-tier supplier networks make digitally enabled GSCMPs especially important (Ghadge et al., 2022).

2.3 Application of Digital Technology in Green Supply Chain Management Practices in the Supply Chain of the Electronics Industry

Industry specifics. The electronics industry has several characteristics that make GSCM both challenging and essential. First, electronic products often have short technology life cycles and fast obsolescence rates, generating large amounts of electronic waste (e-waste) without effective recovery and recycling strategies (Baldé et al., 2024; WHO, 2024). Second, the complex structure of electronic products, including chemical components and toxic metals, requires strict environmental management from production to disposal (WHO, 2024). Third, electronics supply chains often involve multi-tiered global supplier networks, making it difficult to control material origins and upstream environmental impacts (Cicerelli & Ravetti, 2024; De Stefano & Montes-Sancho, 2024). Therefore, GSCMPs such as GP and RL are particularly important for improving the environmental performance of electronics supply chains (Oyedijo et al., 2024; Zhu & Sarkis, 2006).

Typical application of digital technology in each GSCMP of the electronics supply chain. In IEM, IoT sensors help monitor electricity, water and emissions in real time, supporting automated operational optimization and energy-consumption reduction (Poyyamozi et al., 2024). For GP, RFID and IoT applications allow real-time inventory and transportation tracking, thereby improving accuracy and reducing redundancy in the supply chain (Soori et al., 2024). In GM, AI and AI-enabled analytics can support production-load optimization, predictive maintenance and lower manufacturing energy intensity (Wamba et al., 2024; Zhong et al., 2025). In RL operations, IoT and barcodes support the recovery and classification of e-waste, contributing to improved recycling and recovery of valuable materials (Soori et al., 2024). Finally, in EC operations, IoT and blockchain-enabled data sharing can make carbon-footprint information more transparent and strengthen green cooperation among partners (Qiao et al., 2023; Wang & Shen, 2025).

Context of Vietnam. Vietnam's electronics industry accounts for a large share of national exports but remains heavily dependent on imported components, foreign-invested production networks and multinational-customer requirements (ILO, 2023; Vietnam Briefing, 2025; Vietnam News, 2025). Domestic supporting-industry capability remains uneven, and many Vietnamese suppliers still provide relatively simple or low-value products rather than high-technology components (MOIT, 2024). At the same time, electronics firms face increasing ESG pressure from multinational customers and export markets. Digitalization is therefore expected to support greener innovation and emission reduction, but its effects depend on technological readiness, data integration and institutional capacity (Han & Wei, 2025; Wang & Shen, 2025). In emerging contexts, firms may need to follow a sequential pathway: first digitalizing the easier-to-standardize internal stage such as IEM, then expanding data integration, and finally applying BD/AI analytics to GP, GM and RL (Akbari & Hopkins, 2022).

Overall, digital technology is penetrating the entire electronic supply chain, promising to reduce emissions and promote a circular economy. However, the efficiency also depends on the technology readiness of the business and the synchronous coordination throughout the chain.

2.4 Benefits and Barriers of Applying Digital Technology to Green Supply Chain Management Practices

Benefits: The application of digital technology in GSCMPs brings multi-dimensional benefits. IoT and AI help reduce costs, improve efficiency and strengthen sustainable reputation. For example, IIoT sensors combined with AI in manufacturing have been reported to reduce product defects, save raw materials and improve profitability (Poyyamozi et al., 2024; Zhong et al., 2025). BD and AI improve demand forecasting, reducing inventory and costs (Hübner et al., 2024). At the same time, real-time data sharing via IoT or blockchain accelerates coordination, transparency of origin and ecological compliance (Qiao et al., 2023; Wang & Shen, 2025). Studies also show a spillover effect: digitalized upstream enterprises can motivate suppliers to reduce emissions (Bian & Luo, 2025; Han & Wei, 2025).

However, there is still a debate. Some studies warn of the “green paradox”: IoT, AI, and data centers consume large amounts of energy, which can have a negative impact on the environment if renewable sources are lacking (Li, 2022; Zhu & Lan, 2023). In addition, the impact results differ according to each stage of the supply chain - upstream or downstream (Guo et al., 2024). Efficiency is only sustainable when tied to dynamic capabilities and data culture, allowing businesses to restructure processes and transform technology into green advantages (Feroz et al., 2023; Le et al., 2024).

Barriers: Businesses applying technology in GSCM face many barriers. First, high costs: IoT, AI, and BD

investments often account for 10–15% of the project budget and incur maintenance costs (Sarin & Srivastava, 2024). Second, lack of digital human resources: more than 1/3 of businesses face difficulties due to a lack of expertise in data analysis and IoT management (Poyyamozhi et al., 2024). Third, the traditional organizational culture makes many companies slow to accept innovation. Fourth, the lack of synchronization of infrastructure and the lack of common data standards between partners makes it costly to integrate (Sarin & Srivastava, 2024). In addition, the policy environment in many emerging countries, including Vietnam, does not have a strong support mechanism for green digitalization (e.g., tax incentives, green credit). Finally, cybersecurity is a major risk: distributed IoT networks are vulnerable to attacks or data leaks, causing financial and reputational losses (Megias et al., 2024).

In conclusion, the intertwining of benefits and barriers suggests that the impact of adopting digital technologies in GSCMPs cannot be understood as a purely technical issue. The following conceptual framework explains how digital technologies, capability building and institutional conditions interact before the research gaps are specified.

2.5 Conceptual Framework: Digital Technologies, Capability Building and Green Supply Chain Management Practice Diffusion

The conceptual framework links four theoretical lenses to explain the observed IEM-led digitalization pattern. Institutional theory explains the external sustainability pressures that encourage electronics firms to adopt GSCMPs, including ESG expectations, export-market requirements and environmental regulations (DiMaggio & Powell, 1983; Nazir et al., 2024; Zhu & Sarkis, 2006). TOE explains why adoption remains uneven: technological readiness, organizational resources and environmental conditions jointly shape the speed and scope of IoT, BD and AI implementation (Akbari & Hopkins, 2022; Sarin & Srivastava, 2024; Tornatzky et al., 1990). RBV explains why digital technologies alone do not automatically generate green outcomes; they must be combined with valuable and difficult-to-imitate resources such as environmental-management routines, data infrastructure, data-governance rules, skilled employees and supplier relationships (Barney, 1991; Ghadge et al., 2022; Qiao et al., 2023). Dynamic capabilities explain how firms sense sustainability requirements, seize digital opportunities and transform these resources into digital-green capabilities that can be repeatedly reconfigured as sustainability requirements change (Wamba et al., 2024; Li & Lin, 2024; Teece, 2007).

Recent digital-green supply-chain research supports this capability-building view. Supply-chain digitization can enhance corporate green innovation through upstream/downstream integration and improved internal supply-chain efficiency (Ma et al., 2024). It can also reduce carbon emissions through digital transformation and green innovation, with supplier-side effects in some contexts (Han & Wei, 2025). However, AI-enabled and digitally enabled sustainability gains depend on data-driven culture, green capability and organizational transformation rather than technology deployment alone (Wamba et al., 2024; Li & Lin, 2024). Based on this logic, IEM often becomes the first digitalized practice because internal environmental monitoring is more controllable, standardized and directly linked to compliance. Once IEM generates reliable environmental data, firms can build data governance and analytics capabilities that support GM, EC, GP and RL, although outward-facing practices require stronger supplier participation, interoperability and lifecycle data.

2.6 Research Gaps

Research on the intersection of GSCM and digital technology has attracted considerable attention worldwide; however, the literature review in Sections 2.1–2.5 indicates several remaining gaps:

First, most evidence focuses on developed or large markets such as China and the U.S., while research from Vietnam and other emerging economies remains scarce. Existing studies mainly describe overall digitalization but rarely explain how IoT, BD and AI affect each specific GSCMP (Akbari & Hopkins, 2022; Qiao et al., 2023).

Second, few studies jointly examine the three technologies IoT, BD and AI across all five GSCMPs, namely IEM, GP, GM, RL and EC. Most studies focus on single aspects, such as IoT in RL or GP in supplier selection (Tsai et al., 2023).

Third, while global research highlights benefits such as cost and ESG improvements and challenges such as costs, skills and cybersecurity, enterprise-level evidence from Vietnamese firms remains limited. Many electronics firms in Vietnam still face constraints related to domestic value added, supporting-industry capability and digital readiness (ILO, 2023; MOIT, 2024; Vietnam Briefing, 2025).

Fourth, Vietnam's electronics sector has distinctive features, including heavy dependence on imported components, low domestic value added and strong ESG pressure from multinational corporations, yet how these features influence IoT, BD and AI adoption remains underexplored.

Finally, although many studies confirm digitalization's positive and spillover effects (Bian & Luo, 2025; Han & Wei, 2025), others warn of rebound effects from energy use or uneven impacts across supply-chain stages (Guo et al., 2024; Li, 2022), suggesting that contextual dependency remains insufficiently understood.

Based on these gaps, this study provides exploratory multi-case evidence through a comparative analysis of two

analytically informative cases in Vietnam’s electronics industry by: (i) examining the level of IoT, BD and AI adoption across each GSCMP; (ii) identifying the benefits and barriers of digital technology implementation in GSCMPs; and (iii) analyzing how industry-specific contexts influence this adoption. The study thereby extends existing theoretical understanding and provides managerial and policy implications for leveraging digital technologies to advance GSCM in emerging economies, while supporting analytical rather than statistical generalization.

3. Methodology

In the context of Vietnam’s electronics industry, where digital technology adoption remains uneven, data systems are fragmented and ESG frameworks are evolving, a qualitative multi-case study was selected as the optimal approach to explore how IoT, BD and AI influence emerging and context-dependent GSCMPs (Eisenhardt, 1989; Yin, 2018). Compared with quantitative surveys, this design enables rich data exploration, discovery of new variables and relationships through reconstructive logic, and systematic within-case and cross-case comparison (Eisenhardt, 1989; Voss et al., 2002). The triangulation of interviews, observations and documents enhances construct validity and credibility, which are advantages that quantitative methods alone rarely achieve (Ellram, 1996; Patton, 2002; Yin, 2018).

3.1 Research Design and Case Selection

A multi-case design was implemented to strengthen external validity through literal and theoretical replication (Eisenhardt, 1989; Yin, 2018). To ensure contextual consistency, both cases represent leading enterprises in Vietnam’s electronics sector, allowing comparison under “similarity in some conditions and contrast in others” (Voss et al., 2016). Each company is treated as an independent case (level 1), while the five GSCMPs (IEM, GP, GM, RL, EC) act as embedded subunits (level 2). This two-level structure enables in-depth within-case analysis and systematic cross-case pattern identification (Yin, 2018).

Purposive theoretical sampling prioritized information-rich, theoretically significant cases rather than statistical generalization (Eisenhardt, 1989; Siggelkow, 2007). Two large electronics enterprises, Case A and Case B, were selected for (i) pioneering digital transformation with sustainability goals and (ii) distinct business models that facilitate theoretical contrast (electronics-telecom logistics vs. ICT-data center equipment). The cases are therefore not claimed to be statistically representative of all Vietnamese electronics firms. Instead, they are analytically representative of two important digital-green transformation positions in the sector: Case A is a more advanced configuration with integrated data infrastructure and wider GSCMP diffusion, while Case B is an intermediate configuration with partial IoT deployment and weaker data integration. This contrast supports theoretical replication by holding the national/sectoral context relatively constant while varying digital maturity and organizational resources.

Compared with the average firm in Vietnam’s electronics sector, both cases are larger and more digitally mature. Recent sectoral evidence shows that electronics exports reached approximately US\$126.5 billion in 2024 and accounted for about one-third of Vietnam’s total export value, while the localization rate of the electronics industry remains only about 5–10%, with many domestic supporting-industry firms still providing simple, low-technology and low-value products (MOIT, 2024; Vietnam News, 2025). Against this background, the two cases were selected as leading and contrasting cases that reveal potential pathways, bottlenecks and boundary conditions for digital-green transformation, rather than as a statistically representative sample of the entire sector. All organizations and participants were anonymized, and member checking was conducted prior to analysis to ensure credibility (Lincoln & Guba, 1985; Stahl & King, 2020). All participants provided informed consent, and participation was strictly voluntary.

Table 1 summarizes the characteristics of the two case studies and the data collected.

Table 1. Overview of the two case studies and data collected

Case	Main Areas	Scale	Number of Interviews	Digital-Green Maturity	Case-Selection Logic
Case A	Electronics and Telecommunications Manufacturing and Services	Large (>5,000 employees)	5	Advanced	Leading case showing ISO-based EMS, integrated digital infrastructure, supplier/logistics connectivity and wider GSCMP diffusion
Case B	Manufacturing electronic equipment, technology solutions	Large (~1,400 employees)	5	Intermediate	Contrasting case showing partial IoT deployment, fragmented data systems and limited diffusion to GP/RL/EC

Note: ISO = International organization for standardization; EMS = Environmental management system; GSCMP = Green supply chain management practice; IoT = Internet of Things; GP = Green procurement; RL = Reverse logistics; EC = Environmental cooperation.

3.2 Data Collection

The study employed multiple evidence sources to improve construct validity: (i) Literature and document review (sustainability reports, digital strategies, press releases, and academic studies) to frame inquiry; (ii) Semi-structured interviews with 10 managers and 2 experts to obtain comparative insights; (iii) Field observations at production sites and data centers to directly verify IoT/BD/AI applications.

Triangulating interviews, observations, and documents ensured data convergence and reduced single-source bias (Voss et al., 2002; Yin, 2018). Semi-structured interviews offered direction and flexibility to explore emerging topics (Magaldi & Berler, 2020; Ruslin et al., 2022).

Sampling frame. Purposive theoretical sampling maximized information value for cross-comparison (Eisenhardt, 1989; Siggelkow, 2007). Participants met criteria: (i) familiarity with digitalization or GSCM activities; (ii) direct involvement in one of the five GSCMPs; and (iii) willingness to share evidence or documents. Key informants were identified through enterprise recommendations and snowball expansion (Lincoln & Guba, 1985).

Data volume and timeframe. Data were collected between July and September 2025, comprising 12 semi-structured interviews in Vietnamese: 10 internal interviews, including five per firm with senior, technical and environmental managers, and two external expert interviews in supply chain management and digital transformation. Sessions lasted 80–120 minutes and included eight onsite interviews and four online interviews. Around 25 documents, including reports, process maps, policies, KPIs and websites, were analyzed to cross-validate interview findings (Voss et al., 2002; Yin, 2018). Data collection continued until informational redundancy was observed and no substantially new themes emerged, indicating theoretical saturation. Appendix A summarizes the interview participants, including their anonymized NVivo codes, organizational affiliations, positions, interview formats and interview durations.

3.3 Data Analysis

All transcripts, field notes, and documents were processed in NVivo 12 using a hybrid deductive-inductive thematic analysis (Braun & Clarke, 2006). The deductive coding structure was derived from the study's 3×5 analytical matrix: three digital technologies (IoT, BD, AI) across five GSCMPs (IEM, GP, GM, RL, EC). This generated 15 initial technology-practice nodes, such as IoT-IEM, BD-GP, AI-GM and IoT-RL. These nodes were used to identify where and how each technology was applied in each green supply-chain practice. The analysis then proceeded in two phases: within-case mapping of applications, benefits and barriers per firm, and cross-case comparison using pattern matching to align observations with theoretical expectations (Eisenhardt, 1989; Yin, 2018).

In the second coding round, benefits were grouped into operational/economic performance, environmental performance, social sustainability and ESG governance/compliance outcomes. Barriers were grouped into operational-financial, technical-organizational and institutional-market barriers. During inductive coding, additional themes emerged, including data-governance capability, cross-functional coordination, employee digital-sustainability skills, supplier participation, regulatory ambiguity, weak domestic green demand, cybersecurity risk and staged digital-green transition.

To enhance coding reliability, two researchers independently coded a subset of interviews and documents and then compared their coding decisions. Discrepancies were resolved through discussion, codebook refinement and clarification of node definitions before the refined codebook was applied to the full dataset. All coding notes, node definitions, evidence sources and cross-case memos were documented as an audit trail to ensure transparency and verifiability (Lincoln & Guba, 1985). Appendix C provides coding examples showing how first-order codes were aggregated into second-order themes and broader analytical dimensions.

Figure 1 illustrates the research process followed in this study.

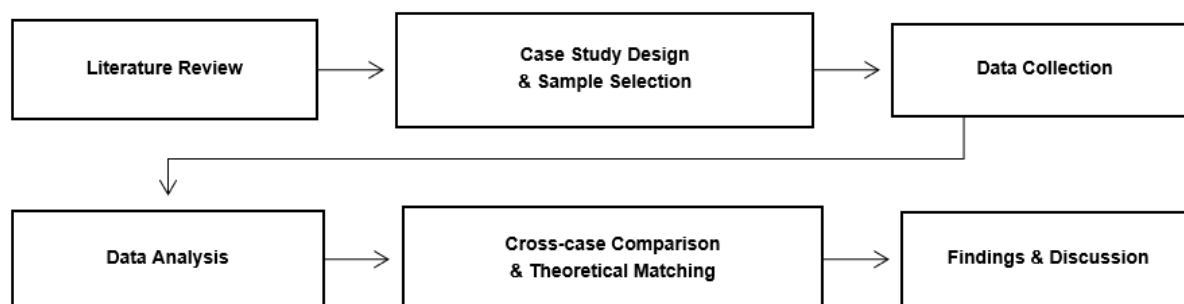


Figure 1. Research process diagram

The research followed six stages: (1) research design and gap identification; (2) case selection and scope definition; (3) data collection; (4) data analysis (coding and within-case mapping); (5) cross-case and theoretical matching; and (6) validation via triangulation and member checking consistent with best practices for multi-case studies (Eisenhardt, 1989; Voss et al., 2016; Yin, 2018).

4. Results and Discussion

The results and discussions are presented according to the main themes, reflecting the proposed approach to matrix analysis. The research team conducted within-case analysis and cross-case comparison based on a 3×5 matrix, including three digital technologies and five GSCMPs. Data from interviews, observations, and documents are encoded, and then matched to clarify similarities and differences (Ridder, 2017). According to Yin (2018), combining multiple sources of evidence enhances credibility, and anonymous citations are used to illustrate.

4.1 Level of Application of Technologies to Green Supply Chain Management Practices

Table 2 below summarizes the results of the analysis according to the 3×5 matrix, comparing the level of IoT, BD and AI application in the five GSCMPs of the two enterprises. The level of application was assessed on a scale from Very High to None, based on NVivo coding and verification through interview data, documentation and field observations. The comparison matrix shows a clear difference between Case A and Case B in terms of maturity in green supply chain digital transformation.

Rating criteria for Table 2. To improve transparency, each application level was assessed using four criteria: deployment breadth, data integration, analytical sophistication and evidence triangulation. Very High indicates broad deployment across relevant operations, real-time data capture, integration with enterprise systems, advanced analytics or automation, and confirmation from at least two evidence sources. High indicates regular deployment with system integration but some limits in automation or practice coverage. Medium indicates partial or function-specific deployment with limited integration. Low indicates pilot, ad hoc or mostly manual use with limited digital evidence. None indicates that no meaningful evidence of application was found in interviews, documents or observations. Hybrid ratings, such as Medium-High, indicate that the evidence falls between two adjacent categories when the application level shows characteristics of both levels.

Table 2. A 3×5 matrix comparing the level of digital-technology application across the five green supply chain management practices (GSCMPs) in Case A and Case B

GSCMPs	Internet of Things (IoT) (A/B)	Big Data (BD) (A/B)	Artificial Intelligence (AI) (A/B)
Internal environmental management (IEM)	Very High/Medium—Hundreds of sensors covering buildings & factories; EMS ISO 14001 provides basic real-time/IoT monitoring at IDC, mostly manual	High/Low—Seasonal EMS analysis & real-time power optimization, HVAC, lighting/No analysis system, mainly manual reporting	Medium-High/None—Intelligent Energy Control, Demand Forecasting/Not yet applied
Green procurement (GP)	Medium-High/None—RFID/IoT Inventory Tracking, Waste Reduction/Not yet applied	High/None—Analysis of consumption history & supplier performance, integration of green criteria/Not yet applied	Medium/None—Demand forecast, automatic purchasing support based on green criteria/Not yet applied
Green manufacturing (GM)	Very High/Medium—Real-time whole-line (Temperature, Vibration) Monitoring/Several Discrete Sensors on SMT, DIP	High/Low—Green production KPIs, real-time dashboard to optimize processes/Discrete data, not yet analyzed in real time	High/Low—Predictive maintenance, load optimization/No AI for maintenance or performance optimization
Reverse logistics (RL)	Medium/Low—IoT & recall tracking barcodes, processing schedules/Manual processes, IoT not yet deployed	High/None—reuse/recycling sorting database, recovery-production data linkage/Unsystematic	Low (developing)/None—Life cycle forecast testing, optimal recall timing/Not yet applied
Environmental cooperation (EC)	High/Low—Sharing of inventory, demand, route data with suppliers & logistics/Limited connectivity testing, lack of standardization	Very High/Low—Shared data platform, 5G connected digital logistics, multi-party real-time access/Discrete exchange, no centralized platform yet	Medium-High/Low—AI & digital twin network optimization, coordination/small-scale pilot

Notes. Ratings were assigned through NVivo coding and triangulated with interview data, company documents and field observations. Very High = broad and integrated deployment with real-time data and advanced analytics/automation; High = regular deployment with partial integration; Medium = function-specific or partial deployment; Low = pilot/ad hoc deployment or limited digital evidence; None = no meaningful evidence identified. Hybrid ratings, such as Medium-High, indicate an intermediate level between two adjacent categories.

Comparative analysis of digital technology application in GSCMP

The comparison suggests an IEM-led diffusion pattern rather than a direct causal spillover. When IEM is deeply digitalized and connected to enterprise data systems, environmental data generated internally can be reused for production optimization, ESG reporting, supplier evaluation and logistics collaboration. We use the term “digitalization spillover” only as an interpretive label for the observed extension of digital-green capabilities from internal environmental monitoring to other GSCMPs. Case A attains Very High IoT-IEM and High BD-IEM integration, extending technologies to GM, RL and EC. Case B, with only Medium/Low IEM, shows more limited diffusion. The largest gaps appear in GP and RL: Case A employs BD and AI to evaluate suppliers, forecast demand and digitize RL, while Case B shows little progress. This interpretation is consistent with recent evidence that supply-chain digitization can support green innovation through internal supply-chain efficiency and upstream/downstream integration, and that carbon-reduction effects may extend to supplier relationships under certain conditions (Han & Wei, 2025; Ma et al., 2024).

IEM. Case A installed hundreds of IoT sensors across facilities, linking to ISO 14001 EMS for real-time monitoring. Dashboards show notable energy savings from waste alerts and early remediation. Conversely, Case B monitors IoT only at IDC, with mostly manual processes, thus limited efficiency. As Case A’s manager noted: “Any abnormal fluctuations are immediately alerted...” (A-TECH1). This is consistent with prior studies reporting that IoT can support energy-use and operational-cost reductions (Poyyamozhi et al., 2024).

GP. Case A embeds environmental criteria into supplier evaluation, uses BD for consumption and performance analysis, and applies RFID/IoT to manage inventory and minimize waste. Case B still relies on standard criteria and experience, without digital tools. This highlights IoT and BD’s contribution to GP, forecasting and logistics optimization, while also showing that GP requires supplier data and shared green standards that are harder to establish than internal monitoring (Ma et al., 2024; Qiao et al., 2023).

GM. Case A shows advanced digitalization: IoT collects real-time production data, with AI adjusting operations, saving energy, and scheduling early maintenance. Case B’s SMT/DIP line lacks predictive AI, relying on fixed schedules. “The system lets machines rest or slow when orders drop and announces maintenance early...” (A-PRO1). These results echo evidence that AI-enabled capabilities and data-driven routines can support environmental performance and reduce energy intensity in manufacturing (Wamba et al., 2024; Zhong et al., 2025).

RL. Case A digitalized the take-back process via IoT and barcodes, ensuring accurate sorting for reuse or recycling. As one environmental manager noted, “IoT helps plan efficient handling for each product type” (A-EM1). Case B’s manual management leads to delayed classification and lower efficiency. This finding is consistent with prior research showing that digital traceability, analytics and database integration can improve recycling efficiency and reverse-logistics coordination (Ma et al., 2024; Qiao et al., 2023).

EC. Case A’s integrated supply chain platform links suppliers and logistics via IoT-5G for real-time flow tracking and route optimization (A-SM1). Shared data reduces latency and emissions. Case B’s tests remain fragmented. This supports evidence that supply chain digitalization fosters green innovation (Wang & Shen, 2025).

Overall, both cases began with IEM, but differences in internal digital capacity led to divergent diffusion outcomes. Case A’s integrated platform enabled wider diffusion across multiple GSCMPs, whereas Case B faced bottlenecks in GP and RL because BD/AI capabilities and data integration remained limited. This pattern is consistent with dynamic capabilities theory, which suggests that firms’ sensing, seizing and transforming capabilities shape their ability to absorb and redeploy digital technologies across practices (Teece, 2007). The findings highlight three key insights: (i) robust internal digital capacity can catalyze operational and environmental efficiency, consistent with evidence on IoT-enabled energy and operational improvements (Poyyamozhi et al., 2024); (ii) BD/AI and data-integration deficits constrain GP–RL diffusion, as shown in the cross-case evidence in Table 2; and (iii) data governance and AI-enabled capabilities help convert technological resources into green and emission-reduction benefits (Wamba et al., 2024; Li & Lin, 2024; Wang & Shen, 2025).

4.2 Benefits of Digital Technologies in Green Supply Chain Management Practices

Based on the iterative multi-case strategy (Yin, 2018), the benefits from the application of IoT, BD and AI in GSCMPs can be generalized into four groups aligned with the journal’s environmental, economic and social sustainability emphasis: (i) Operational and Economic Performance (OP), (ii) Environmental Performance (ENV), (iii) Social Sustainability (SOC), and (iv) ESG Governance, Brand Reputation and Compliance (BR/CMP). NVivo analysis shows that OP is the most mentioned, ENV is prominent in Case A but less developed in Case B, SOC emerges mainly through employee capability development and supplier participation, and BR/CMP increases rapidly in Case A thanks to real-time ESG data.

OP. Economic and operational benefits are the most frequently reported benefit category. Case A applies IoT line monitoring, real-time control panels and load-optimized AI, associated with a reported reduction of approximately 20% downtime, shorter lead time and a reported logistics-cost reduction of 30–40% through the integrated IoT-5G-Digital Twin logistics platform (A-PRO1; A-SM1). In contrast, Case B achieves more moderate efficiency gains: IoT mainly serves IDC management, BD is applied only in inventory, and RL remains manual

(B-SM1; B-PRO1). This disparity may be because Case A has already deployed an integrated digital logistics platform, while Case B is still at the early stage of digitizing warehousing. This finding is consistent with prior evidence showing that digital technology can improve network design, shorten cycles, reduce disruption time and strengthen supply-chain competitiveness (Akbari & Hopkins, 2022; Qiao et al., 2023).

ENV. The difference is more pronounced in environmental performance. Case A deploys AI to optimize the load and hibernation mode of the 5G network, with a reported saving of approximately 200 million kWh/year (A-TECH1), and digitizes RL with reported reuse of 7-8 out of 10 recovered devices (A-EM1). In contrast, Case B is low-to-medium: IoT manages IDC, ISO 50001 certification supports electricity saving, but RL is still manual with a low recycling rate (B-EM1; B-SM2). The disparity partly stems from the scale of investment and the level of green technology application of Case A, allowing Case A to achieve greater energy savings and a higher reuse rate than Case B. This observation is consistent with recent studies indicating that supply-chain digitization and AI can help reduce energy consumption, cut carbon emissions and improve environmental performance when supported by green innovation and organizational capabilities (Han & Wei, 2025; Wamba et al., 2024; Zhong et al., 2025).

SOC. The social-sustainability evidence in this study is more limited than the operational and environmental evidence. Some aspects are directly supported by the case evidence, particularly employee capability development, skill shortages, supplier data participation and safer monitoring practices. The cases indicate that employees require training in IoT operation, data analytics, environmental monitoring and cybersecurity awareness (A-TECH1; B-TECH1; B-PRO1). Supplier participation is also directly reflected in Case A's data-sharing arrangements and procurement-related green criteria, and in Case B's more limited connectivity with partners (A-SM1; A-PR1; B-SM2). Other aspects, such as accountable data-sharing arrangements and broader community implications, are interpreted more cautiously as implications derived from the cross-case analysis and the social-sustainability literature. This is relevant in Vietnam's electronics industry, where the ILO highlights decent-work and skills-related challenges in the electronics supply chain (ILO, 2023), and where digitalized circular supply chains can create social and governance tensions related to data privacy, power asymmetry and fairness (Zighan & Ruel, 2025).

ESG Governance, BR/CMP. The ESG governance and compliance benefits are most evident in Case A. Thanks to real-time ESG data, Case A makes reporting more transparent, responds quickly to policy changes, and strengthens its green business positioning: "When regulations change, we can reproduce the report in hours instead of weeks" (A-SM1). Case B is only average, relying heavily on ISO 14001 certification and small cooperation initiatives (B-SM1). These observations are consistent with empirical evidence that GSCM practices improve corporate image, increase customer trust, and reduce the risk of violating the law (Zhu et al., 2019), and are also compatible with institutional theories of compliance pressure and social expectations (DiMaggio & Powell, 1983).

Overall, Case A appears to attain stronger operational and economic performance, environmental performance, emerging social-sustainability benefits and ESG governance/compliance benefits than Case B, while Case B exhibits more limited SOC evidence and average BR/CMP. Based on triangulated case evidence, we interpret two value mechanisms cautiously rather than treating them as direct interview terms or quantitatively validated causal effects. First, real-time ESG data appear to create a compliance-flexibility mechanism: firms can respond more quickly when reporting requirements change because sustainability data have already been captured, structured and made auditable. We use the term compliance option value only as an interpretive label for this flexibility under regulatory uncertainty. Second, digitalized reverse-logistics and reuse data suggest a potential asset-lifecycle extension mechanism. In Case A, recovered devices and components are digitally classified and reused where technically feasible, which may reduce the immediate need for replacement purchases. This remains an interpretation requiring further quantitative validation to estimate its financial magnitude (Gu, 2025).

4.3 Barriers to Implementing Digital Technologies in Green Supply Chain Management Practices

Despite its benefits, applying IoT, BD & AI in GSCMPs still encounters significant barriers in both Case A and Case B, grouped into three clusters: (i) Operational & Financial Barriers (OFB), (ii) Technical & Organizational Barriers (TOB), and (iii) Institutional & Market Barriers (IMB).

OFB. Financial limitations remain the top challenge. Case B emphasized cost pressure, with a manager admitting: "Upgrading data infrastructure is the biggest challenge today" (B-SM1). Reliance on internal budgets and absence of external incentives make B's burden heavier. Case A also acknowledged high initial IoT/AI costs but mitigated them through pilot-scale deployment and leveraging its existing NB-IoT infrastructure (A-SM1). The "pilot first, expand later" strategy optimized cash flow and avoided large upfront shocks (A-PRO1). This aligns with previous findings that GSCM digitization often consumes 10-15% of project budgets and may exceed estimates (Chouaibi et al., 2022; Sarin & Srivastava, 2024).

TOB. The shortage of technological and GSCM skills is a major barrier. Case B struggles due to a lack of IoT/BD/AI experts, while Case A has partly resolved this via recruitment, training, and establishing a dedicated unit (A-TECH1). This reflects a broader deficiency in "data thinking" across firms (Poyyamozhi et al., 2024), highlighting the urgency of developing digital human resources (Sarin & Srivastava, 2024).

Data and system integration further complicate progress. Case A overcame earlier issues by adopting middleware and API/data standards, effectively linking IoT-ERP systems. Case B, however, faces delays in building its data lake due to fragmented legacy systems (B-SM2), a common bottleneck among firms lacking uniform data standards (Sarin & Srivastava, 2024). Cybersecurity and privacy risks are also crucial. Case A treats each IoT device as a potential weak point, applying end-to-end encryption, multi-factor authentication, and 24/7 monitoring. Case B piloted blockchain for data integrity but admitted to “limited employee safety awareness” (B-PRO1), echoing NIST warnings (Megas et al., 2024) that breaches can cause severe data and reputational losses.

IMB. Two Vietnam-specific barriers are (i) unclear ESG regulations and (ii) weak domestic green demand. Case A noted: “Each agency uses different criteria; we spend time translating the same ESG data” (A-SM1). Case B also faced delays due to vague guidelines (B-SM1). Moreover, local customers remain highly price-sensitive, prioritizing cost over sustainability (A-SM1; B-SM2). This finding extends the TOE framework by emphasizing the influence of domestic policy and market demand on the speed of GSCM digitalization (Nazir et al., 2024).

Overall, Case B faces stronger financial, skill, integration and institutional barriers, whereas Case A, although still challenged, shows higher efficiency through pilot IoT use, middleware and real-time ESG monitoring. These obstacles form a “vicious circle”: limited digital skills → weak integration → higher risks → slow ROI → reduced investment motivation. In particular, vague ESG regulation and weak domestic green demand are Vietnam-specific constraints that obscure the benefits of digitalization.

When juxtaposed with the benefits (Section 4.2), the cases suggest a digital-green capability-building pattern: Case A has climbed the green digitalization value ladder by digitizing IEM, integrating data, applying AI in selected GP/RL activities and developing real-time ESG reporting, while Case B remains intermediate with an incomplete data lake. Three interpretive mechanisms stand out rather than general causal effects: (1) real-time ESG data may offer compliance flexibility amid regulatory ambiguity; (2) reuse of RL equipment may reduce or defer some investment costs; and (3) Vietnam’s distinctive ESG and market barriers appear to shape the speed of digital-green diffusion. Table 3 provides an evidence map that links these interpretive claims to interview codes, document evidence and cross-case interpretation.

Table 3. Evidence mapping for key interpretive claims

Key Claim	Supporting Evidence in this Study	Evidence Type/Source	Interpretation Status
Supplier participation	Case A’s supplier/logistics data sharing and procurement-related green criteria; Case B’s limited connectivity and standardization.	Interview codes A-SM1, A-PR1 and B-SM2; company documents; Table 2 EC/GP evidence.	Direct case evidence plus cross-case interpretation.
Cybersecurity risk	Case A’s end-to-end encryption, MFA and monitoring; Case B’s reported limited employee safety awareness.	Interview codes A-TECH1 and B-PRO1; IT/security documents.	Direct case evidence.
Compliance flexibility	Case A’s statement that reports can be reproduced quickly when regulations change.	Interview code A-SM1; ESG/reporting documents.	Direct evidence, interpreted cautiously as compliance flexibility.
Potential asset-lifecycle extension	Digitally classified recovered devices and reuse tracking in reverse logistics.	Interview code A-EM1; reverse-logistics/reuse documents.	Direct evidence; financial mechanism remains interpretive.
SME boundary conditions	Large-firm case selection and sectoral evidence on low localization and uneven digital capability.	MOIT, Vietnam Briefing and Vietnam News sources; Section 3.1 case-selection rationale.	Contextual/literature-supported boundary condition, not direct SME evidence.
Social sustainability	Employee training, skill gaps, supplier data participation and safer environmental monitoring.	Interview codes A-TECH1, B-TECH1, B-PRO1, A-SM1 and A-PR1; ILO and Zighan & Ruel.	Mixed: direct case evidence plus literature-supported implication.

Note: SME = Small and medium-sized enterprises; MFA = Multi-factor authentication.

4.4 Comparison of Green Supply Chain Management Digitization Strategy Between Case A and Case B

On the basis of the differences in the presented results, this section analyzes and contrasts the strategic orientation of digitalization associated with the greening of the supply chain of two typical electronics enterprises in Vietnam, following a replication logic in multi-case study research (Eisenhardt, 1989; Ridder, 2017).

Technology priorities. Case A prioritizes IoT, 5G, BD and AI to optimize logistics and whole-chain operations, achieving reported operational and environmental efficiency improvements, including an approximately 20% downtime reduction and substantial electricity savings reported by interviewees and company evidence. Case B focuses on smart factories based on industrial IoT and AI to improve productivity and inventory management, but the spillover effect to RL and EC is limited.

Ecosystem expansion. Case A builds a domestic logistics ecosystem through a data platform connecting firms, customs and transportation partners, thereby increasing transparency and compliance. Case B expands through

international joint ventures and receives advanced technology, but the internal green benefits remain less clear because it lacks a full-chain integrated data platform.

Organizational policies and culture. Case A focuses on energy efficiency, renewable energy, and green building standards, and strengthens data governance to respond to volatile environmental policies. Case B invests in high-tech human resources and applies ISO certification, but is hampered by high costs, limited human resources and unclear institutions.

The results show that while both companies pursue digitalization and greening goals in parallel, they have significant differences in their strategic focuses, technology roadmaps, and how to overcome barriers in the institutional and domestic market contexts.

Cross-case explanatory mechanisms. The difference between Case A and Case B is not explained only by firm size or investment level. Three organizational mechanisms explain why digital technologies diffuse more extensively across GSCMPs in Case A: digital-green capabilities, data-governance mechanisms and leadership commitment. First, Case A combines IoT infrastructure, environmental-management routines, analytics capability and cross-functional knowledge across production, procurement, logistics and environmental management. Second, Case A has more mature data governance, using middleware, API/data standards and integrated platforms to connect environmental, production, logistics and supplier-related information. Third, Case A treats digitalization and sustainability as linked strategic priorities supported by pilot-and-scale routines, whereas Case B remains more constrained by fragmented legacy systems, limited analytics, budget dependence and weaker inter-functional integration. This is consistent with the view that digital transformation improves environmental performance only when mediated by green capability, data-driven culture and organizational transformation (Wamba et al., 2024; Li & Lin, 2024). Table 4 summarizes the cross-case explanation of digital-green diffusion by comparing digital-green capability, data governance, leadership commitment, cross-functional coordination and supplier/partner participation across the two cases.

Table 4. Cross-case explanation of digital-green diffusion

Explanatory Mechanism	Case A	Case B	Implication for GSCMP Diffusion
Digital-green capability	Integrated IoT, BD/AI, EMS and logistics data capabilities	Function-specific IoT and early-stage analytics	Case A diffuses digitalization beyond IEM; Case B remains concentrated in factory/IDC applications
Data governance	Middleware, API/data standards and integrated platforms	Fragmented legacy systems and slower data-lake development	Stronger data governance enables diffusion to GM, EC, GP and RL
Leadership commitment	Digitalization and sustainability treated as linked strategic priorities	Digitalization more cautious and resource-constrained	Stronger commitment supports scaling beyond pilots
Cross-functional coordination	Environmental, production, technology and logistics functions connected	Less integrated coordination across functions	Coordination supports evidence reuse and faster compliance response
Supplier/partner participation	More developed supplier/logistics data sharing	Limited connectivity and standardization	EC, GP and RL depend on inter-organizational data participation

Note: GSCMP = Green supply chain management practice; GP = Green procurement; RL = Reverse logistics; EC = Environmental cooperation; GM = Green manufacturing.

In summary, the within-case and cross-case analysis using the 3×5 matrix suggests that the diffusion mechanism depends on the maturity of IEM and the level of data integration. When IEM is deeply digitalized and data are interconnected, technologies may extend to GM, EC and RL; conversely, the absence of BD/AI creates bottlenecks in GP and RL due to real-time requirements for traceability and green evaluation. The differences between Case A and Case B primarily stem from system integration, data governance, cross-functional coordination and leadership commitment rather than investment scale alone, while the boundary conditions of non-standardized ESG regulations and weak green demand slow down diffusion. These findings are consistent with dynamic capabilities theory and suggest that data governance may serve as a converting capability that helps transform technological resources into improved GSCM performance.

4.5 Digital-Green Capability-Building Roadmap and Policy Implications

Building on the case evidence and cross-case interpretation, the findings can be translated into a five-step digital-green capability-building roadmap. Figure 2 presents the roadmap, while Table 5 summarizes the evidence base for each step. The roadmap is presented as a managerial and policy implication rather than as a general causal model, because it is derived from exploratory multi-case evidence.

The roadmap follows five linked steps. Step 1, Digitalize IEM, focuses on standardizing environmental

processes, such as ISO 14001/50001, and deploying IoT sensors for energy and environmental monitoring. This creates a reliable data baseline. Step 2, Build integrated data governance, links IoT, MES and ERP systems via middleware and unified data governance, forming a digital backbone that can reduce system fragmentation. Step 3, Apply BD/AI analytics, leverages centralized data for demand forecasting, green supplier assessment, optimized production and predictive maintenance, thereby supporting efficiency and environmental performance. Step 4, Extend digital collaboration, expands digitalization across the supply chain by establishing shared data spaces, batch/lot traceability and greater supply-chain transparency with key partners. Step 5, Strengthen cybersecurity and accountability, safeguards this digital-green infrastructure through encryption, multi-factor authentication, continuous monitoring, IoT/database security and clear cross-functional accountability.

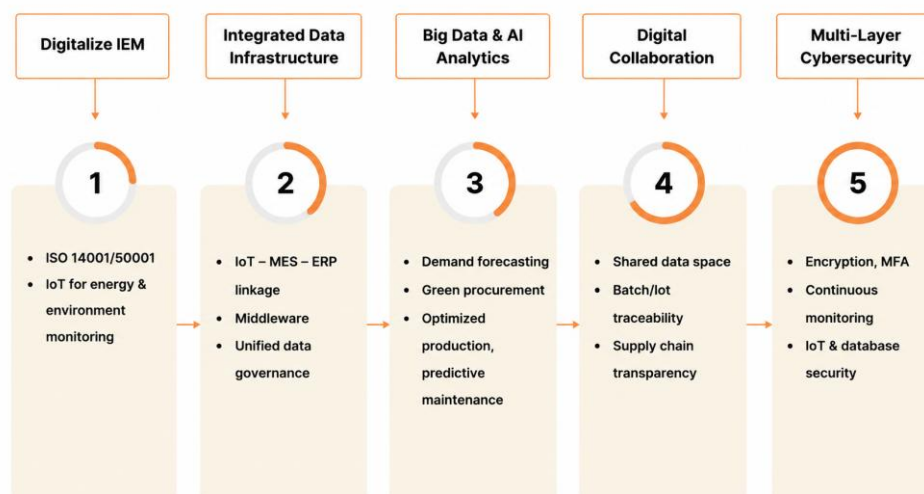


Figure 2. Digital-green capability-building roadmap for green supply chain management practice (GSCMP) diffusion

Table 5. Evidence base for the proposed roadmap

Roadmap Step	Evidence from Cases	Interpretation
Step 1. Digitalize IEM	Both cases began with internal environmental or energy monitoring; Case A developed stronger IoT-enabled EMS	IEM is the most controllable entry point for digital-green transformation
Step 2. Build integrated data governance	Case A connected systems through platforms and standards; Case B faced fragmented systems	Data governance helps convert digital tools into scalable GSCMP capabilities
Step 3. Apply BD/AI analytics	Case A used analytics for forecasting, maintenance and supplier-related evaluation; Case B remained limited	Analytics enable transition from monitoring to optimization
Step 4. Extend digital collaboration	Case A shared data with logistics/supply-chain partners; Case B had limited connectivity	GSCMP diffusion requires inter-organizational participation
Step 5. Strengthen cybersecurity and accountability	Both cases recognized cybersecurity risk as IoT connectivity expanded	Digital-green transformation requires governance safeguards

Note: BD = Big data; IEM = Internal environmental monitoring; EMS = Environmental management system; GSCMP = Green supply chain management practice.

Policy implications. To accelerate digital technology adoption in GSCM across emerging economies such as Vietnam, the findings point to the need for a coherent policy package that reduces structural bottlenecks and investment risks at both firm and national levels. Key priorities include: (i) strengthening national ESG governance by issuing clearer ESG reporting guidelines, standardized environmental indicators and consistent data requirements; (ii) expanding financial incentives through green credit lines, innovation funds and tax relief for IoT/AI projects in energy management, reverse logistics and cleaner production; (iii) standardizing Industry 4.0 data protocols and environmental data formats for energy, emissions, waste, supplier traceability and reverse logistics to enhance interoperability; (iv) linking digital transformation policy with green industrial upgrading so that IoT, BD and AI programmes explicitly support cleaner production, e-waste governance and supply-chain transparency; and (v) building digital-green capabilities for SMEs through shared digital platforms, targeted training, technical advisory programmes and sandbox pilots. Such policies would improve regulatory predictability, strengthen national ESG systems and support Vietnam's wider industrial transformation toward sustainable electronics supply chains.

5. Conclusions

Key findings. Based on exploratory multi-case evidence from two leading Vietnamese electronics firms, the findings map the application level of IoT, BD and AI across the five GSCMPs, synthesize the benefits and barriers of digitalizing GSCMPs, and explain how the context of Vietnam's electronics industry shapes adoption. The findings support analytical rather than statistical generalization. To reduce repetition, the conclusion synthesizes the findings around the observed IEM-led digitalization pattern, data-governance capability and Vietnam-specific boundary conditions rather than restating each research question separately.

Synthesis of the IEM-led digitalization pattern. The analysis shows marked variability in digitalization across GSCMPs. IEM is the most digitalized practice because it is internal, compliance-oriented and easier to standardize; both firms deploy IoT-based monitoring, while the more advanced case integrates BD and AI for proactive environmental control. GM and EC become more digitalized when IEM data are connected to production, logistics and collaboration systems. By contrast, GP and RL remain less developed because they require supplier data, lifecycle information, interoperability and stronger analytics. Overall, the cases suggest an observed IEM-led digitalization pattern: robust internal environmental data systems may enable wider diffusion to GM and EC, whereas weak internal capabilities leave outward-facing practices such as GP and RL under-digitalized. This pattern requires further validation across broader samples and other organizational contexts.

Synthesis of benefits, barriers and contextual conditions. The cases show that IoT, BD and AI can generate operational/economic value, environmental performance improvements, emerging social-sustainability benefits and stronger ESG governance/compliance. At the same time, adoption is constrained by high investment costs, shortages of digital and GSCM skills, fragmented data systems, cybersecurity risks, unclear ESG standards and weak domestic green demand. The cases suggest two emerging interpretive mechanisms: compliance flexibility generated by real-time ESG data and potential asset-lifecycle extension through digitally supported reuse. These mechanisms require further quantitative validation. The cross-case comparison further indicates that adoption success depends less on investment size alone than on data governance, cross-functional coordination, leadership commitment and policy clarity.

Theoretical contributions. The study contributes to theory in three exploratory ways. First, based on two analytically informative Vietnamese electronics cases, it integrates institutional theory, TOE, RBV and dynamic capabilities into a digital-green capability-building framework. Institutional theory explains sustainability pressure, TOE explains adoption conditions, RBV explains digital-green resources, and dynamic capabilities explain how firms may transform these resources into GSCMP improvements. Second, the study suggests data governance as a converting capability that may help turn IoT, BD and AI resources into practice-specific improvements in IEM, GM, EC, GP and RL. Third, the study proposes an IEM-led digitalization pattern as an observed diffusion mechanism that should be tested further across broader samples, SME contexts and other emerging economies.

Limitations and future research. This study has several limitations. First, it focuses on two large electronics firms in one emerging economy; therefore, the findings support analytical rather than statistical generalization and should not be interpreted as representative of all Vietnamese electronics firms or all emerging-economy supply chains. The boundary condition is especially important for SMEs with weaker financial, digital, data-governance and ESG-reporting capabilities. For smaller firms, the IEM-led digitalization pattern may need to be more modular, starting with low-cost environmental monitoring, basic data standardization, shared platforms and external training before moving to BD/AI analytics or inter-organizational data sharing. Second, the cross-sectional design cannot fully capture how digital-green capabilities evolve over time; longitudinal studies are needed. Third, the study focuses on IoT, BD and AI, leaving technologies such as blockchain, digital twins and robotics for future research. Further studies should also quantify compliance flexibility and asset-lifecycle extension mechanisms under different regulatory, SME-capability and supply-chain governance conditions.

Author Contributions

Conceptualization, D.-N.N. and T.-T.M.; methodology, D.-N.N. and T.-T.M.; investigation (including data collection from interviews, documents, and observations), T.-T.M., D.-N.N., T.-H.T., M.-A.T., and H.-H.H.; resources, H.-H.H.; data curation, T.-T.M. and M.-A.T.; formal analysis, T.-T.M. and D.-N.N.; validation, D.-N.N. and H.-H.H.; visualization, T.-T.M., M.-A.T., and H.-H.H.; supervision, T.-T.M. and D.-N.N.; technical supervision, H.-H.H.; project administration, D.-N.N. and T.-T.M.; writing original draft preparation, T.-T.M. and D.-N.N.; writing review and editing, D.-N.N., T.-T.M., T.-H.T., M.-A.T., and H.-H.H. All authors have read and agreed to the submitted version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all participants involved in the study. Participation was voluntary, and all organizations and interviewees were anonymized.

Ethical Approval

This study involved voluntary interviews with adult professional participants in managerial, technical and environmental roles. No sensitive personal, medical or vulnerable-participant data were collected. According to the authors' institutional guidelines, formal institutional ethical approval was not required for this minimal-risk, non-medical and anonymized interview-based study. The research was conducted in accordance with ethical research principles, including voluntary participation, informed consent, anonymity, confidentiality and the right to withdraw.

Data Availability

The raw interview transcripts, interview recordings, field notes and company documents cannot be publicly disclosed because they contain confidential business information and were collected under anonymity and confidentiality agreements. Aggregated and anonymized evidence is provided in the article and Appendices A–C. Additional anonymized coding summaries or aggregated evidence may be made available from the corresponding author upon reasonable request, subject to participant consent, company confidentiality restrictions and institutional data-protection requirements.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Appendix A.

Table A1. List of interview participants

NVivo Code in Analytics	Company	Position	Interview Format	Time (Minutes)
A-SM1	Case A	Senior Management	Offline	80
A-EM1	Case A	Environmental Management	Online (Teams)	85
A-TECH1	Case A	Technology Management	Offline	90
A-PRO1	Case A	Production Management	Offline	120
A-PR1	Case A	Procurement Management	Online (Teams)	80
B-SM1	Case B	Senior Management	Offline	95
B-SM2	Case B	Senior Management	Online (Teams)	100
B-EM1	Case B	Environmental Management	Offline	90
B-PRO1	Case B	Production Management	Offline	115
B-TECH1	Case B	Technical Management	Online (Teams)	90
EXP1	Expert	Independent GSCM Expert	Offline	120
EXP2	Expert	Supply Chain Digital Transformation Advisor	Offline	110

Appendix B.

Table B1. Summary of the benefits and barriers to green supply chain management (GSCM) digitalization: Comparing Case A and Case B

Group	Aspect	Case A	Case B	Compare
Benefit (4.2)	Operational Performance (OP)	Very high: IoT line monitoring, AI maintenance forecasting showed a reported reduction of approximately 20% downtime; digital logistics is associated with a reported logistics-cost reduction of 30-40% High: AI optimizes loads; 5G “hibernation” mode showed a reported saving of approximately 200 million kWh/year; RL digitalization reported reuse of 7-8 out of 10 recovered devices	Medium: New IoT adopted at IDC; BD supports limited inventory reduction; RL is still manual	Case A much higher than Case B
	Environmental Performance (ENV)		Low-Medium: IDC (ISO 50001) energy-saving IoT management; Manual RL, low reuse rate	Case A much higher than Case B
	Brand Reputation & Compliance (BR/CMP)	Medium-High: Real-time ESG application transparency reporting, positioning “green business”	Medium: Maintain ISO 14001, “green IDC” signals, and technology cooperation to improve images	Case A moderately higher than Case B
Barrier (4.3)	Operational & Financial Barriers	High: The investment cost is large but there are resources; reduce the burden by small	Very high: IoT/BD/AI/GSCM costs are too large, budget is	Case B much higher than

(OFB)	pilots and leverage NB-IoT	limited, corporation dependence, lack of incentives	Case A
Technical & Organizational Barriers (TOB)—Skills & Human Resources	Medium: Initially lacking but adding experts, internal training, IoT team formation	High: Serious shortage of IoT/BD/AI/GSCM personnel; training has not kept up with demand	Case B much higher than Case A
Technical & TOB—System & Data Integration	Medium: The technology system is still legacy but fixed with middleware, protocol standards	High: Complex discrete systems; Data lakes and data standardization progress slowly	Case B much higher than Case A
TOB—Cybersecurity & Privacy	Medium: High risk but good control: E2E encryption, MFA, 24/7 monitoring; No major incidents have occurred	Medium-High: Expanding IoT increases risk; Testing blockchain but employee awareness is still weak	Case B much higher than Case A
Institutional & Market Barriers (IMB)—Regulatory Ambiguity	Medium: ESG regulations are inconsistent, must be reported many times; real-time ESG data create compliance flexibility	High: ESG guidelines are unknown, many green projects are slow to approve	Case B much higher than Case A
IMB—Weak Domestic Green Demand	Medium: Domestic customers are price-sensitive; Green pressure mainly from exports	Medium: Domestic customers/partners focus on price & progress; faint green element	Case A and Case B broadly similar
Social Sustainability (SOC)	Emerging: employee training, data-skills development, supplier participation and safer environmental monitoring	Limited: skill shortages and weaker supplier data participation constrain social dimension	Case A higher than Case B

Note: “much higher than” indicates a substantial qualitative difference between the two cases; “higher than” or “moderately higher than” indicates a moderate qualitative difference; and “broadly similar” indicates comparable evidence across the two cases.

Appendix C.

Table C1. Coding examples used in NVivo thematic analysis

Aggregate Dimension	Second-Order Theme	First-Order Code	Illustrative Evidence	Evidence Source
Technology-practice application	IoT-enabled IEM	Real-time energy/environmental monitoring	Sensors and dashboards used to detect abnormal fluctuations in energy or environmental indicators	Interview, observation, EMS documents
Technology-practice application	BD-enabled GP	Supplier and consumption analytics	Consumption history and supplier performance data used to support green purchasing decisions	Interview, procurement documents
Technology-practice application	AI-enabled GM	Predictive maintenance and load optimization	AI-supported scheduling reduces downtime and adjusts equipment operation according to production load	Interview, production documents
Digital-green capability	Data governance	Middleware, API standards, integrated data lake	Integrated data structure enables IEM data to support manufacturing, reporting and collaboration	Interview, IT documents
Sustainability outcome	Environmental performance	Energy saving, waste reduction, reuse	Digital monitoring and reuse tracking support energy reduction and component recovery	Interview, company documents
Sustainability outcome	Social sustainability	Employee capability and supplier participation	Staff training and supplier data sharing are required to scale digital-green practices	Interview, training/cooperation documents
Institutional condition	ESG ambiguity	Multiple reporting criteria	Firms spend additional time translating similar ESG data into different reporting formats	Interview, ESG/reporting documents
Barrier	Cybersecurity risk	IoT device vulnerability	More connected devices increase data-security and monitoring requirements	Interview, IT/security documents